MAN (Metropolitan Area Network): A MAN covers a larger area than a LAN, usually a whole city or a large campus. For example, a network that connects different buildings across a university campus in one city is a MAN.

WAN (Wide Area Network): A WAN covers a very large area, often across cities, countries, or even the whole world. The internet itself is the biggest example of a WAN, since it connects networks from all over the world together.

PAN (Personal Area Network): A PAN is a very small network, usually just around one person, connecting their personal devices. For example, when a phone connects to a headset using Bluetooth, that is a PAN.

Wired vs Wireless Networks

Wired	Wireless
Faster and more secure	Flexible and portable
Uses Ethernet cables	Uses Wi-Fi and Bluetooth
Less interference	More prone to interference

Wired Networks: A wired network connects devices using physical cables, usually Ethernet cables. Because the connection is physical, it is generally faster and more secure than a wireless connection. It also has less interference, since there is no signal passing through the air. However, it is less flexible, because devices must stay connected to the cable.

Wireless Networks: A wireless network connects devices without cables, using technologies like Wi-Fi or Bluetooth. This makes it more flexible and portable, since devices can move around freely without being plugged in. However, wireless networks are more prone to interference, from things like walls, other electronic devices, or other wireless signals nearby.

Ethernet, Cables & Fiber Optics —
Networking Basics Every IT Professional Should Know

Ethernet Standards & Speeds (IEEE 802.3)

- 10BASE-T - 10 Mbps (802.3I)
- 100BASE-T - 100 Mbps (802.3u)
- 1000BASE-T - 1 Gbps (802.3ab)
- 10GBASE-T - 10 Gbps (802.3an)

Maximum Distance (Copper Ethernet)

- Up to **100 meters** using UTP cables with RJ-45 connectors



Cable Types & Usage

Straight-Through Cable

- Used for: Computer → Switch
- Router → Switch

Crossover Cable

- Used for: Computer → Computer
- Switch → Switch



Ethernet Pin Assignment



- Transmit (TX): Pins 1 & 2
- Receive (RX): Pins 3 & 6

Auto MDI-X

- Modern network devices automatically adjust TX/RX RX pins – no need to worry about cable type anymore



Bottom line:

- Choose UTP for cost-effective short runs and Fiber for high speed, long distance, and secure networks.

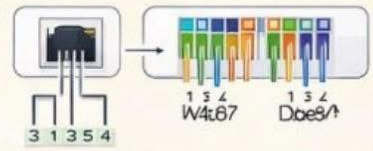
Which medium do you prefer

Maximum Distance (Copper)



- Transmit (TX): Pins 1 & 2
- Receive (RX): Pins 3 & 6

Ethernet Pin Assignment



- Transmit (TX): Pins 1 & 2
- Receive (RX): Pins 3 & 6

Auto MDI-X

- Modern network devices automatically adjust TX/RX pins – no need to worry about cable type anymore

Fiber Optics — Speed, Distance & Security

Single-Mode Fiber (SMF)

- Laser-based
- Long distance (up to 30 km)



Multi-Mode Fiber (MMF)

- LED-based
- Short distance (up to 550 meters)



UTP vs Fiber Optic – Quick Comparison

	UTP	Fiber
Cost	Lower	Higher
• Distance	• 100 meters	• Up to 30 km
• EMI Resistance	• Prone to interference	• 100% immune
• Security	• Possible signal leakage	• No signal leakage (more secure)

Access Point (AP)

An Access Point, usually called an AP, is a device that changes a wired network into a wireless network. In simple words, it takes the internet connection coming through a cable and spreads it out as Wi-Fi, so that devices nearby can connect without needing a cable.

Let's understand why this is needed with an example. Imagine a large university with hundreds of students and many buildings. If we only had one Wi-Fi router in the main office, its signal would not be strong enough to reach every classroom, hallway, and building. Also, one router cannot handle that many devices connecting at the same time.

So, to solve this problem, the university does the following:

- There is one main router, which is the central point where the internet enters the building.
- From this main router, long network cables are pulled out to different parts of the campus.
- At the end of these cables, Access Points are installed in various locations, such as on walls, ceilings, or light poles, so that they are spread evenly across the area.
- The Access Point receives the internet through the wire connected to it, but then it broadcasts that internet wirelessly, so that laptops and phones nearby can connect to Wi-Fi.

This way, instead of depending on one single router, the university can cover a very large area with strong Wi-Fi signals, by using many Access Points connected through cables to the main network.

The OSI Model (7 Layers)

The OSI Model is a conceptual framework that explains how data travels from one computer to another computer across a network. It breaks down this whole process into 7 separate layers, where each layer has its own specific job. This makes it easier to understand, design, and troubleshoot networks, because we can focus on one layer at a time instead of the whole complicated process at once.

Here is what each layer does:

1. Physical Layer

This is the lowest layer, and it deals with the actual physical hardware of the network. Its job is to transmit raw bits (0s and 1s) over physical media, such as cables or fiber optic lines. Examples of technologies at this layer include Ethernet cables and fiber optics.

2. Data Link Layer

This layer makes sure that data is transferred correctly, without errors, between two devices that are on the same local network. It handles things like detecting and fixing errors that may happen during transmission. Examples at this layer include Ethernet and Wi-Fi.

3. Network Layer

This layer is responsible for deciding the best physical path that data should take to reach its destination, especially when data needs to travel across different networks. The main protocols used here are IPv4 and IPv6, which handle addressing and routing of data.

4. Transport Layer

This layer manages end-to-end communication between two devices. It takes large chunks of data and breaks them into smaller segments, so they can be sent more easily. It also makes sure that these segments arrive correctly and in the right order. The two main protocols here are TCP and UDP.

5. Session Layer
This layer manages the connection, or "session," between two computers. It is responsible for opening, maintaining, and closing communication sessions. Examples of protocols at this layer include RPC and PPTP.

6. Presentation Layer
This layer acts like a translator. It makes sure the data is in a format that both computers can understand. It is also responsible for handling encryption, so data can be transmitted securely. Examples at this layer include JPEG (for images) and SSL/TLS (for secure connections).

7. Application Layer
This is the topmost layer, and it is the layer where users actually interact with the network. This is where familiar applications and services work, such as web browsing, sending emails, and transferring files. Examples of protocols here include HTTP (for websites), FTP (for file transfer), and SMTP (for email).

Why do we use the OSI 7-layer model?

The OSI model was developed as a conceptual way to understand and explain how communication happens from one point in a network to another point. It helps engineers and students understand networking step by step, layer by layer.

In practical, real-world usage, however, we usually simplify these 7 layers into just 4 layers. This simplified version is called the TCP/IP model, and it is what is actually used in real network systems today.

Protocols

A network protocol is a set of established rules that controls how data is formatted, transmitted, and received between devices on a network. Without protocols, devices would not be able to understand each other.

To understand this better, let's think about it like a phone call between two people:

- First, you say "Hello."
- Then, the other person also says "Hello."
- If you then start speaking in Sinhala, the other person must also understand Sinhala for the conversation to continue.
- This pattern, saying "Hello" first and then speaking in the same language, is like a protocol.

If either person breaks these rules, for example, if you speak in Sinhala but the other person only understands Chinese, then the communication completely fails, and no useful information is exchanged.

In the same way, for different types of devices on a network, such as laptops, servers, and routers, to communicate with each other successfully, they need to follow a common set of rules that everyone agrees to use. This common set of rules is what we call a protocol.

Ports

A port is a logical connection point on a computer. It acts like a gateway that allows specific types of data to enter or leave the computer.

The main purpose of a port is to help the operating system tell the difference between different types of network traffic. For example, web traffic, email traffic, and gaming traffic all use different ports, so the computer knows exactly where to send each type of incoming data, and which application should handle it.

TCP vs UDP

Both TCP and UDP are protocols used at the Transport Layer, and they are both used to send data across a network, but they work in very different ways.

TCP (Transmission Control Protocol)

TCP is a very reliable method of sending data. Before any data is actually sent, a proper connection is first established between the two devices that want to communicate.

Here is how it works: Each piece of data is given a sequence number, so the receiving side knows the correct order to put them back together. When the receiving device gets a piece of data, it sends back an acknowledgement message, basically saying "I received this piece successfully." If any piece of data gets lost along the way, the sender will notice, and it will resend that missing piece.

Because of this reliability, TCP is used for tasks where accuracy is very important, such as:

- Web browsing (HTTP/HTTPS)
- Sending emails
- File transfers (FTP)

UDP (User Datagram Protocol)

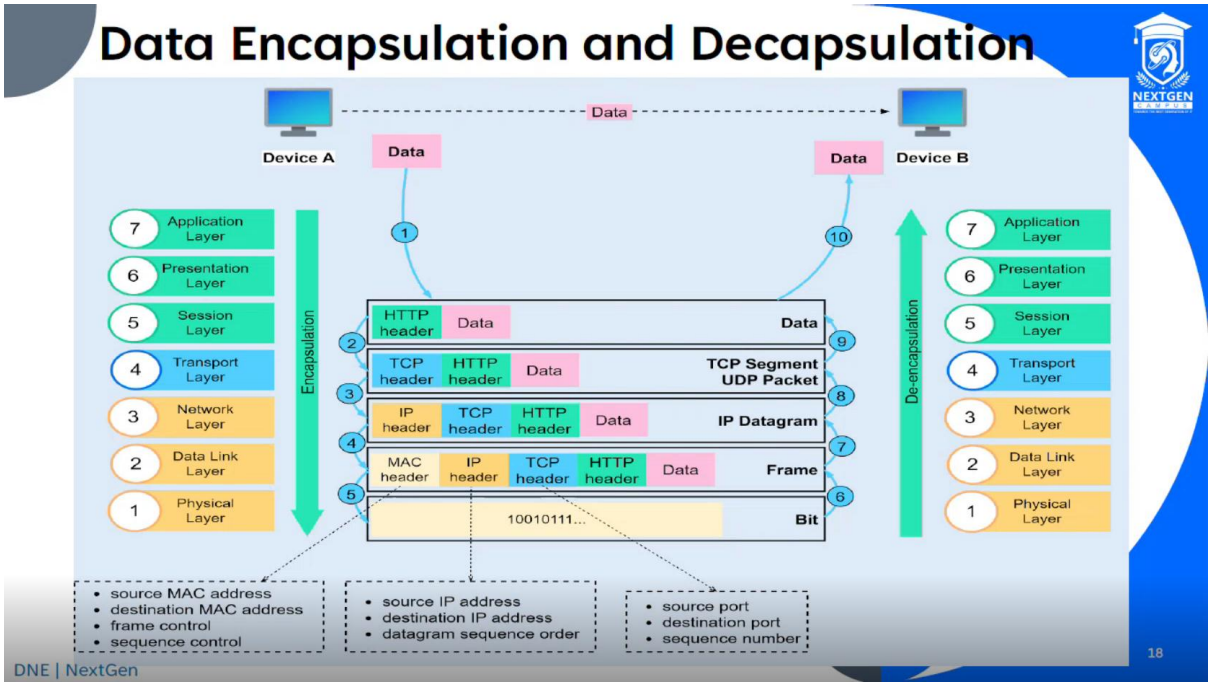
UDP, on the other hand, is a very fast method of sending data, but it does not check whether the data has actually been received by the other side. This is why it is called "connectionless."

Here is how it works: Data is simply sent out continuously, without waiting for any confirmation. There is no "I received it" message sent back from the receiving side. If some data gets lost during transmission, it is simply not resent, and the receiving side just continues on without that missing piece.

Because UDP prioritizes speed over reliability, it is used for tasks where a small amount of lost data is acceptable, but delay is not, such as:

- Video streaming (like YouTube Live)
- Online gaming
- Voice calls (like WhatsApp calls, also known as VoIP)
- DNS lookups

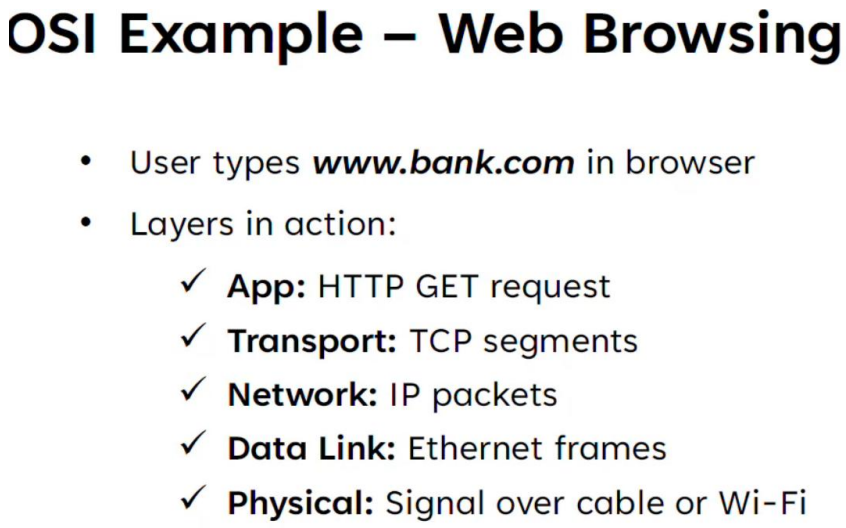
How work OSI model



Why add HTTP headers, TCP headers and others to data via transferring?

To transferring data correctly and accurately

OSI Example: Web Browsing



Let's understand how the OSI model actually works in real life, using a simple example: a user types www.bank.com into a web browser.

When this happens, several layers of the OSI model work together, one after another, to make this request possible:

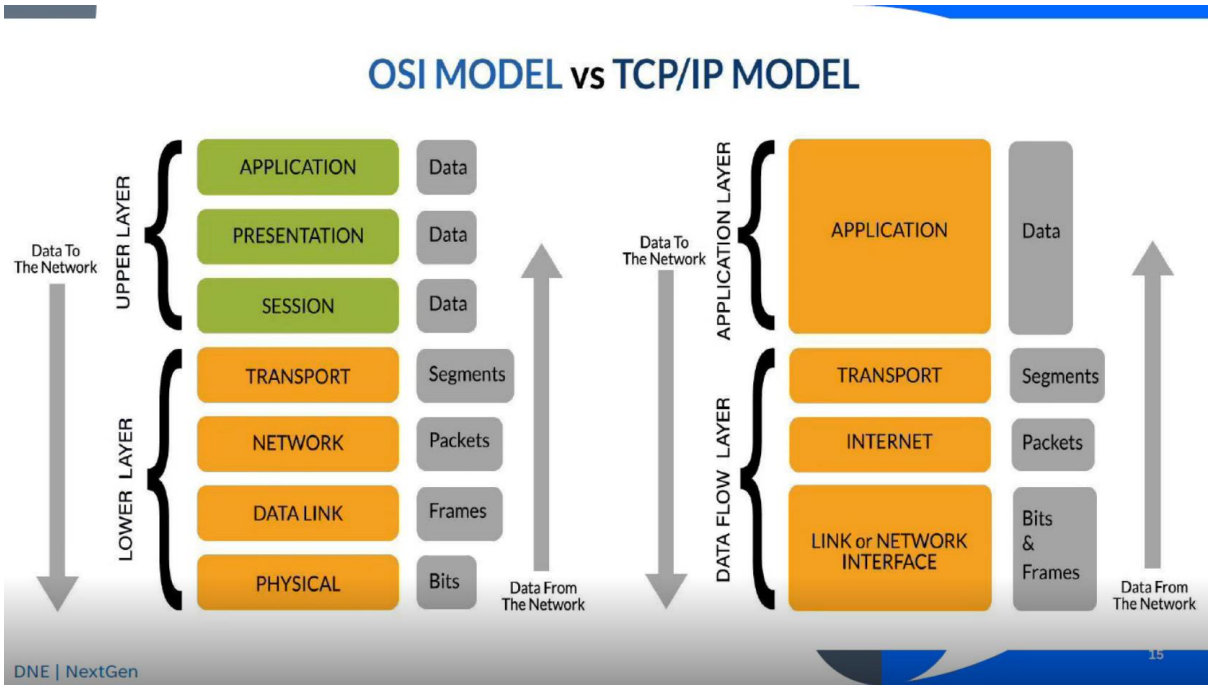
- **Application Layer:** The browser creates an HTTP GET request. This is basically the browser's way of asking the bank's server, "Please send me your webpage."
- **Transport Layer:** This request is broken down into TCP segments. TCP makes sure the data will arrive completely and in the correct order.

- **Network Layer:** The data is placed into IP packets. These packets contain addressing information, so the data knows where to go.
- **Data Link Layer:** The IP packets are placed inside Ethernet frames, so they can travel correctly across the local network.
- **Physical Layer:** Finally, all of this data becomes an actual electrical or radio signal, which travels through a cable or through Wi-Fi.

This example shows that even a simple action, like visiting a website, actually involves all 7 layers of the OSI model working together in the background.

OSI Model vs TCP/IP Model

The OSI Model has 7 layers, but in real-world practical use, we simplify this into the TCP/IP Model, which only has 4 layers. Here is how they compare:



OSI Model (7 Layers)	TCP/IP Model (4 Layers)
Application, Presentation, Session	Application
Transport	Transport
Network	Internet
Data Link, Physical	Link or Network Interface

In the OSI model, the top 3 layers (Application, Presentation, and Session) are combined into just one single layer, called the Application layer, in the TCP/IP model. Similarly, the bottom 2 layers (Data Link and Physical) are combined into one layer, called the Link or Network Interface layer.

This shows that the TCP/IP model takes the same overall idea as OSI, but groups the layers together in a simpler, more practical way for real-world use.

Key Differences: OSI vs TCP/IP

Feature	OSI Model	TCP/IP Model
Number of Layers	7	4
Developed By	A conceptual model created by ISO (International Organization for Standardization)	A practical model created by DARPA
Protocol Dependency	Protocol-independent, meaning it is just a general concept, not tied to specific protocols	Protocol-based, meaning it is built directly around the TCP/IP protocols
Layer Segmentation	More detailed and granular, since it has more layers	Fewer, broader layers, which combine multiple functions together
Main Use	Mainly used for teaching and understanding networking concepts	Mainly used in real-world, practical implementation of networks

Why Do We Add HTTP Headers, TCP Headers, and Other Headers to Data?

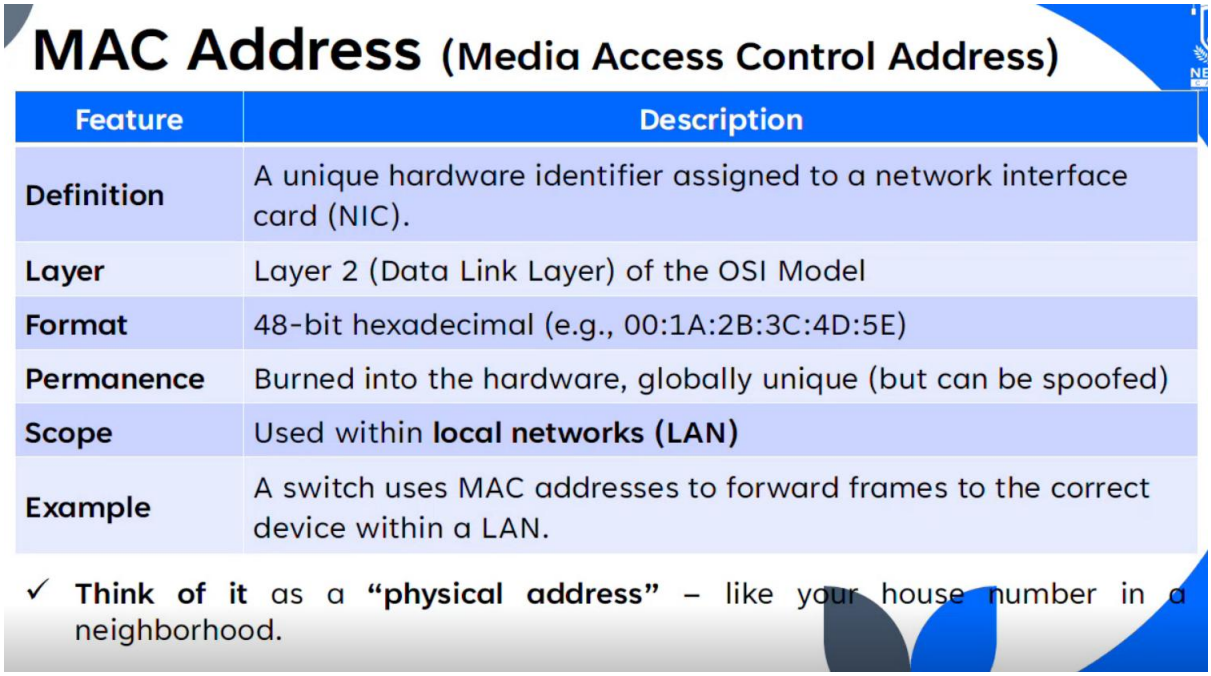
Every time data is sent across a network, extra information called "headers" is added to it, at different layers. The reason we do this is to make sure the data is transferred correctly and accurately.

These headers contain important details, such as where the data is coming from, where it needs to go, how it should be organized, and how to check if it arrived without errors. Without these headers, the receiving device would not know how to correctly read, organize, or verify the data that it receives.

MAC Address (Media Access Control Address)

A MAC address, also called a physical address, is a unique identifying name given to a piece of hardware by the hardware company that made it. More specifically, a MAC address is a unique hardware identifier that is assigned to a Network Interface Controller (NIC), by the manufacturer of that device.

Every network device, such as a laptop's Wi-Fi card, a network card, or a switch port, has its own unique MAC address, which is given to it during manufacturing.



Which layer does it work in?

The MAC address works at Layer 2, which is the Data Link Layer, of the OSI Model. This is the layer responsible for making sure data is transferred correctly between two devices on the same local network.

What format does it use?

A MAC address is written as a 48-bit hexadecimal number. For example: 00:1A:2B:3C:4D:5E. It is usually written as six pairs of characters, separated by colons or dashes.

Is it permanent?

Normally, the MAC address is burned into the hardware during manufacturing, and it is meant to be globally unique, meaning no two devices should have the exact same MAC address. However, it is technically possible to "spoof" or change the MAC address using software, even though it is originally fixed.

Where is it used?

A MAC address is used within local networks (LAN). It is not used for communication across the wider internet, that is handled by IP addresses instead.

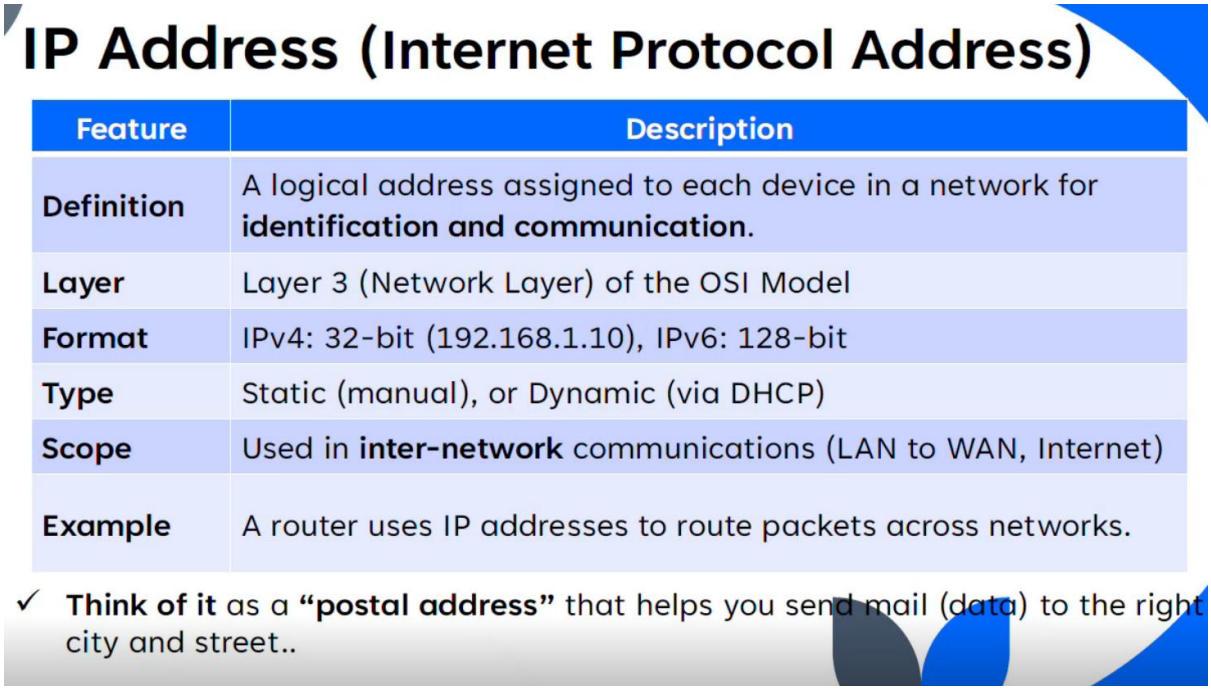
Example of how it's used:

A switch uses MAC addresses to correctly forward data frames to the right device within a LAN. This means when data arrives at a switch, the switch checks the MAC address to know exactly which device on the network the data is meant for.

Simple way to remember it:

We can think of a MAC address like a "physical address," similar to a house number in a neighborhood. Just like every house has its own unique number so mail can be delivered correctly, every network device has its own unique MAC address so data can be delivered to the correct device.

IP Address (Internet Protocol Address)



An IP address is a logical address that is assigned to each device in a network. Its main purpose is identification and communication, meaning it helps identify a device on a network, and it allows that device to send and receive data.

Which layer does it work in?

An IP address works at Layer 3, which is the Network Layer, of the OSI Model. This is the layer responsible for deciding the best path for data to travel across networks.

What format does it use?

There are two versions of IP addresses:

- **IPv4:** This is a 32-bit address, written in a format like 192.168.1.10.
- **IPv6:** This is a newer, 128-bit address, created because the world was running out of available IPv4 addresses.

How is an IP address assigned?

An IP address can be assigned in two ways:

- **Static:** The IP address is set manually, and it does not change unless someone changes it again.
- **Dynamic:** The IP address is assigned automatically, using a service called DHCP (Dynamic Host Configuration Protocol).

Where is it used?

Unlike a MAC address, which is only used within a local network, an IP address is used for inter-network communication. This means it is used when data travels between different networks, for example, from a LAN to a WAN, or across the internet.

Example of how it's used:

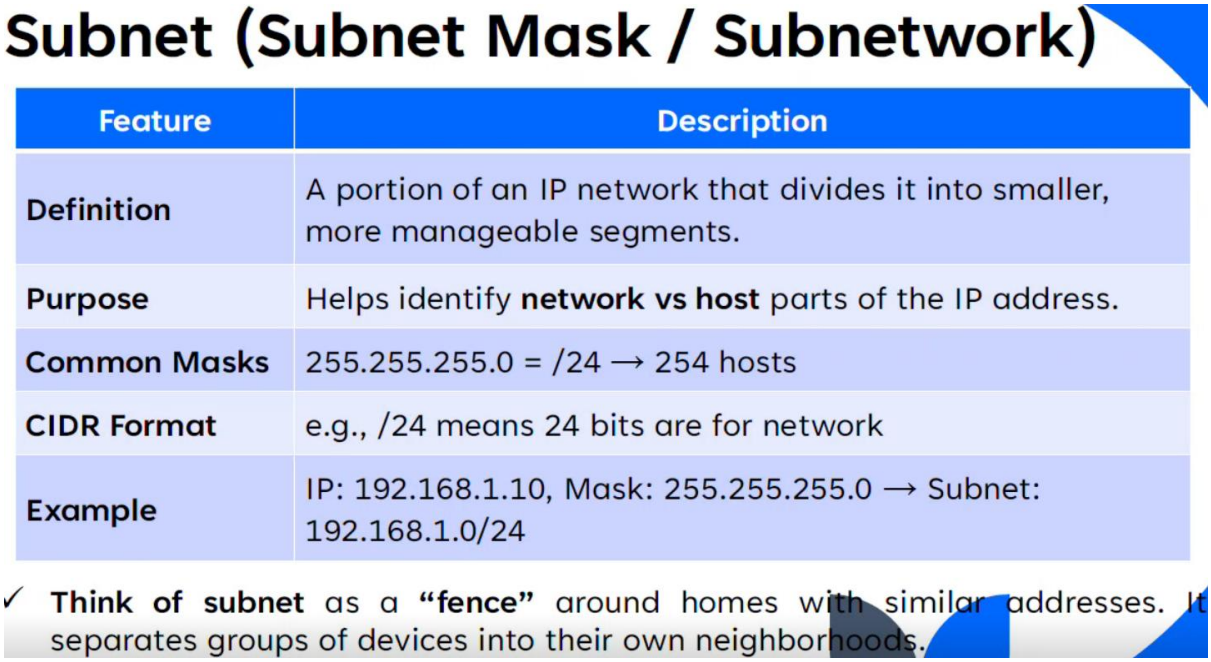
A router uses IP addresses to correctly route data packets across different networks. This means when data needs to travel from one network to another, the router looks at the IP address to decide the correct path to send it.

Simple way to remember it:

We can think of an IP address like a "postal address." Just like a postal address helps deliver mail (data) to the correct city and street, an IP address helps deliver data to the correct device across different networks.

Subnet (Subnet Mask / Subnetwork)

A subnet is a portion of a larger IP network, which is divided into smaller, more manageable segments. Instead of having one huge network with thousands of devices all mixed together, a subnet lets us break that network down into smaller, organized sections.



What is the purpose of a subnet?

The main purpose of a subnet mask is to help identify which part of an IP address represents the network, and which part represents the host (the individual device). Every IP address is actually made up of these two parts, and the subnet mask tells us exactly where the network part ends and the host part begins.

Common subnet masks:

One common example is the subnet mask 255.255.255.0, which is also written in CIDR format as /24. This particular mask allows for 254 usable host addresses within that subnet.

What is CIDR format?

CIDR stands for a shorthand way of writing subnet masks. For example, /24 means that 24 bits of the IP address are reserved for the network part, and the remaining bits are used for the host part (individual devices).

Example:

If we have an IP address of 192.168.1.10, with a subnet mask of 255.255.255.0, then the subnet is written as 192.168.1.0/24. This means all devices with IP addresses starting from 192.168.1.1 to 192.168.1.254 belong to this same subnet.

Simple way to remember it:

We can think of a subnet like a "fence" built around a group of homes that have similar addresses. This fence separates different groups of devices into their own neighborhoods, keeping each group organized and manageable, even though they might all still belong to the same larger city (the overall network).

Example: MAC Address, IP Address, and Subnet Mask

Device	MAC Address	IP Address	Subnet Mask
PC1	00:11:22:33:44:55	192.168.1.10	255.255.255.0
PC2	00:11:22:33:44:99	192.168.1.20	255.255.255.0
Router	00:11:22:33:44:AA	192.168.1.1	255.255.255.0

Scenario 1: PC1 sends data to PC2

Since PC1 and PC2 are both on the same subnet (192.168.1.0/24), PC1 does not need to go through the router. Instead, PC1 uses a process called ARP to find PC2's MAC address, so it can deliver the data directly.

Scenario 2: PC1 sends data to 192.168.2.10

In this case, the destination address (192.168.2.10) is outside PC1's own subnet. This means PC1 cannot directly reach the destination, so it must send the data through the router instead, and the router will forward it to the correct network.

ARP Table (Address Resolution Protocol Table)

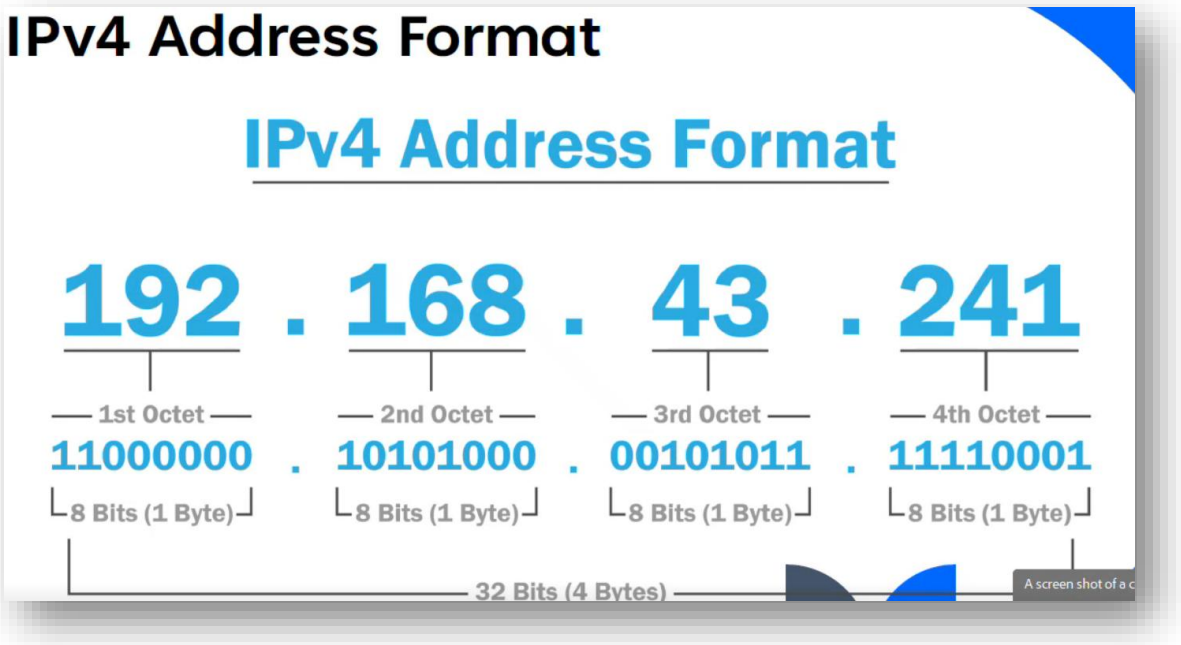
An ARP Table is used to map IP addresses to their corresponding MAC addresses. Devices like PCs, routers, and Layer 3 switches all use this table.

The purpose of the ARP Table is simple: whenever a device only knows the logical address (IP address) of another device, but needs to know the physical hardware address (MAC address) to actually deliver the data, it uses ARP to look this up.

For example, when PC1 wants to send data to PC2, PC1 already knows PC2's IP address, but it does not automatically know PC2's MAC address. So, PC1 uses the ARP process to ask, "Who has this IP address? Please tell me your MAC address." Once PC2 responds with its MAC address, PC1 stores this information in its ARP table, so it can deliver data directly next time, without needing to ask again.

IP Addressing

IPv4 Address Format



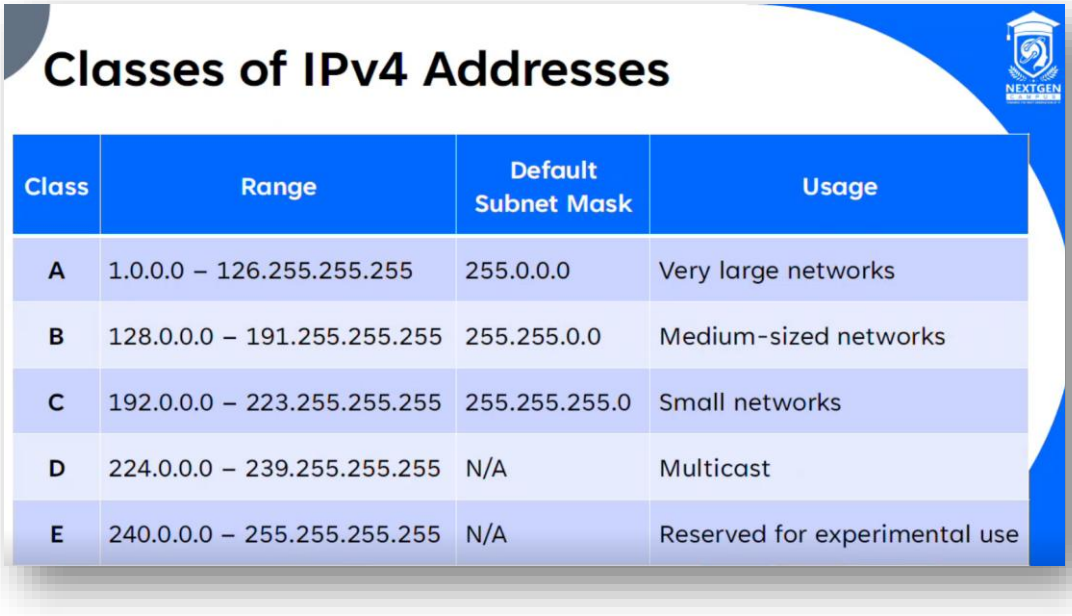
An IPv4 address is made up of 32 bits in total, which is equal to 4 bytes. It is written as four numbers separated by dots, and each of these four numbers is called an "octet," because each one represents 8 bits (1 byte).

For example, in the IP address 192.168.43.241:

- The 1st octet is 192, which in binary is 11000000
- The 2nd octet is 168, which in binary is 10101000
- The 3rd octet is 43, which in binary is 00101011
- The 4th octet is 241, which in binary is 11110001

When we add up all four octets together, we get 32 bits in total (4 bytes), which makes up the complete IPv4 address. Computers actually read and process this address in binary form, but humans usually read and write it in the simpler, dotted decimal format (like 192.168.43.241), since that is much easier for us to understand and remember.

Classes of IPv4 Addresses



IPv4 addresses are divided into different classes, and each class is meant for a different size or type of network:

Class	Range	Default Subnet Mask	Usage
A	1.0.0.0 – 126.255.255.255	255.0.0.0	Very large networks
B	128.0.0.0 – 191.255.255.255	255.255.0.0	Medium-sized networks
C	192.0.0.0 – 223.255.255.255	255.255.255.0	Small networks
D	224.0.0.0 – 239.255.255.255	N/A	Multicast
E	240.0.0.0 – 255.255.255.255	N/A	Reserved for experimental use

Class A is used for very large networks, because it allows for a huge number of individual host devices within a single network. This class is typically used by very large organizations or internet service providers.

Class B is used for medium-sized networks, which is smaller than Class A, but still supports a large number of devices. This class is typically used by universities or larger companies.

Class C is used for small networks, and it is the most commonly used class for everyday networks, like home networks or small offices, since it supports a smaller, more manageable number of devices.

Class D is not used for normal devices at all. Instead, it is reserved for multicast, which means sending the same data to multiple devices at the same time, such as for streaming or video conferencing.

Class E is reserved for experimental and research purposes, and it is not used for regular networking at all.

NAT (Network Address Translation)

Why do we use NAT? What is NAT?

The total number of available IPv4 addresses in the whole world is limited, only about 4.3 billion addresses in total. However, the actual number of devices in the world, like phones, laptops, smart TVs, and more, is much larger than that.

Let's think of a simple example. Imagine we have 10 devices in our home or office. If we wanted to give each of these 10 devices its own separate public IP address, it would be very expensive, and we would quickly run out of available public IP addresses, since there simply aren't enough for every single device in the world.

Solution: This is where NAT comes in. NAT allows all 10 of those devices to share and access the internet using just a single public IP address, instead of needing 10 separate ones.

How does NAT actually work?

The router is responsible for doing this translation work. Here is how it works, step by step:

- **Outbound (when data leaves our network):** When our laptop, for example with a private IP of 192.168.1.10, sends a request to a website like Google, the router removes our private IP address from the data packet, and instead attaches its own public IP address to it, since that is the address that can actually be recognized on the internet.
- **NAT Table:** At the same time, the router keeps a written record in something called the NAT Table. This record basically says, "This particular request came from a device called 192.168.1.10 inside our network," so that later, it remembers exactly where to send the reply back to.
- **Inbound (when data comes back into our network):** When the response data comes back from Google, it arrives at the router's public IP address first. The router then checks its NAT table, sees which internal device originally made the request, and forwards that data back to the correct device, in this case, our laptop.

This is how one single public IP address can be shared by many different devices at the same time, without any confusion about which device asked for what.

How to Share Files Between Computers on the Same Network

If we have multiple computers connected to the same network, and we want them to share files with each other, we need to follow a few steps.

Step 1: Check IP Addresses

All the computers need to be on the same subnet, so that they are able to "talk" to each other properly.

- To check this, we open Command Prompt (CMD) on each computer, and type the command ipconfig.
- All the computers should have IP addresses that start with the same numbers, for example, all starting with 192.168.1.X.
- The subnet mask should also be the same on all devices, usually 255.255.255.0.

Step 2: Set a Static IP (Recommended)

If the main computer, meaning the one that is sharing its files, has its IP address change unexpectedly, then the other devices will lose their connection to it. This is why it is a good idea to give the main computer a fixed, static IP address instead.

Here's how to do that:

1. Go to Control Panel, then Network and Sharing Center, then Change adapter settings.
2. Right-click on the network connection being used, and select Properties.
3. Select IPv4 from the list, and click Properties again.
4. Enter a fixed IP address, for example 192.168.1.50.

Step 3: Enable Network Discovery

By default, computers on a network are sometimes not able to "see" each other. We need to give them permission to do this.

1. Search for "Advanced sharing settings" using the Windows search bar.
2. Turn ON the "Network discovery" option.
3. Turn ON the "File and printer sharing" option.
4. Optionally, under "All Networks," we can turn OFF "Password protected sharing," if we want to share files without needing to type a password every single time.

Step 4: Share the Folder

1. Right-click on the folder we want to share, and select Properties.
2. Go to the "Sharing" tab, and click "Advanced Sharing."
3. Check the box that says "Share this folder."
4. Click on "Permissions," and choose whether to give "Everyone" Read access (only allows viewing) or Change access (allows editing as well).

Step 5: Access the Shared Files from Other Devices

Now, from any of the other computers on the same network:

1. Open File Explorer.
2. In the top address bar, type two backslashes \\, followed by the IP address of the main computer. For example: \\192.168.1.50
3. Press Enter, and the shared folder should now appear, ready to be accessed.

Class	Range	Default Subnet Mask
A	1.0.0.0 – 126.255.255.255	255.0.0.0
B	128.0.0.0 – 191.255.255.255	255.255.0.0

127?

Why Don't We Use the 127.x.x.x IP Range?

Looking at the table, we can see that the 127.x.x.x range sits right between Class A and Class B. However, this specific range is not actually used for normal devices on a network. Instead, it is reserved for something special, called the loopback range.

What is a Loopback Address?

A loopback IP address is a special address that a computer uses to talk to itself. When we use this address, the data packet does not actually go out onto the network, meaning it does not pass through a switch or a router. Instead, the data simply "loops" back through the computer's own network card, straight back into the same computer.

Even though this range technically belongs to the Class A address system, it is generally reserved specifically for this loopback purpose, and not used for regular devices:

- **Main address:** 127.0.0.1, which is very commonly known as "localhost."
- **Range:** If we try to ping any address between 127.0.0.1 and 127.255.255.254, for example ping 127.0.0.2, we will actually get a reply back. This happens because this entire range is reserved for the device to talk to itself, no matter which specific address within that range we use.

Within this range:

- 127.0.0.0 is considered the network address.
- 127.255.255.255 is considered the broadcast address.

Understanding Network ID and Broadcast Address

Every IP network has two special addresses that are not given out to actual devices:

- **Network ID:** This is the address where all the host bits are set to 0. This address represents the network itself, not any specific device within it.
- **Broadcast Address:** This is the address where all the host bits are set to 1. This special address is used to send data to every single device on that network at the same time.

How the Subnet Mask Decides Network vs Host Bits

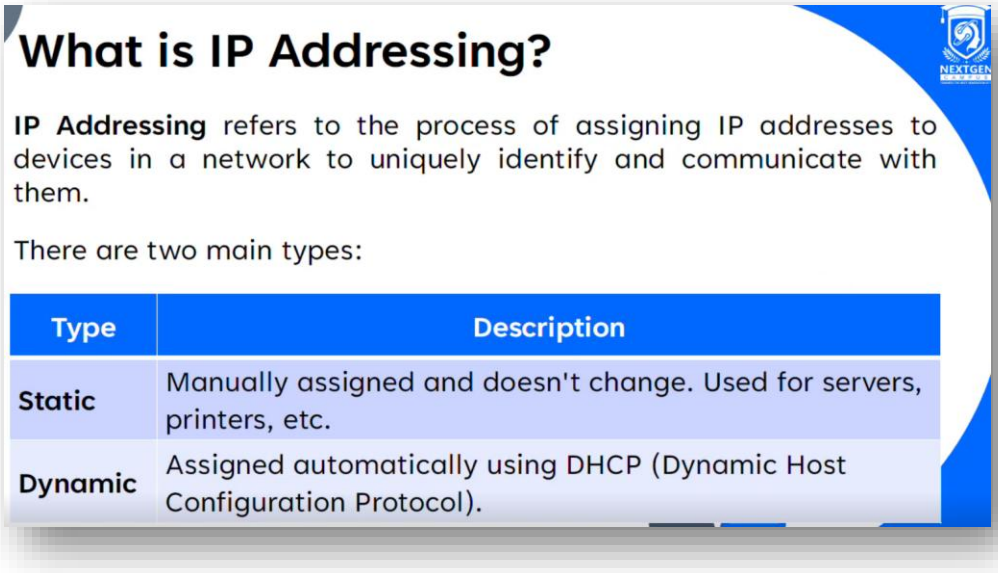
Depending on which subnet mask we use, different parts of the IP address are considered the "network" part versus the "host" part:

- If the subnet mask is 255.255.255.0, then the first three octets represent the Network, and only the last octet represents the Host. (Network.Network.Network.Host)
- If the subnet mask is 255.255.0.0, then the first two octets represent the Network, and the last two octets represent the Host. (Network.Network.Host.Host)
- If the subnet mask is 255.0.0.0, then only the first octet represents the Network, and the last three octets represent the Host. (Network.Host.Host.Host)

This shows us that depending on the subnet mask being used, the same IP address can be divided differently between its network portion and its host portion

What is IP Addressing?

IP Addressing refers to the process of assigning IP addresses to devices in a network, so that each device can be uniquely identified and can communicate with others.



There are two main types of IP addressing:

Type	Description
Static	The IP address is assigned manually, and it does not change. This is usually used for servers, printers, and other devices that need a fixed address.
Dynamic	The IP address is assigned automatically, using a service called DHCP (Dynamic Host Configuration Protocol).

Why Do We Use Subnetting?

1. Identifying Network and Host Portions

Subnetting helps us figure out which part of an IP address belongs to the network, and which part belongs to the host (the individual device).

2. Determining the Network Boundary

Subnetting helps a device decide whether a destination IP address is on the same local network, or whether the data needs to be sent through a gateway (router) to reach a different network.

3. Dividing Networks (Subnetting)

Subnetting also determines the maximum number of devices (hosts) that are allowed to connect within a specific network. For example, using a /24 subnet mask allows for up to 254 devices on that network.

Important Example: Can These Devices Communicate?

Let's say we have 4 devices, with the following IP addresses, and all of them are using the same subnet mask, 255.0.0.0:

- Device A: 192.168.1.1
- Device B: 172.28.1.1
- Device C: 192.168.1.2
- Device D: 10.10.10.1

Question: Can all of these devices ping each other, using only a switch (no router)?

Answer: No, they cannot all communicate with each other.

Why not?

Even though all the devices are using the same subnet mask, that alone does not mean they are on the same network. We also need to look at the actual network portion of their IP addresses:

- 192.x.x.x is a different network than 172.x.x.x
- 192.x.x.x is a different network than 10.x.x.x

Since the subnet mask is 255.0.0.0, only the first 8 bits of the IP address represent the network, and the remaining bits represent the host. Because these devices have different numbers in their very first octet (192, 172, and 10), they actually belong to completely different networks, even though their subnet mask is the same.

Because of this, without a router to connect these different networks, only Device A and Device C can communicate with each other, since they are both on the same 192.168.x.x network. Device B (172.x.x.x) and Device D (10.x.x.x) are on separate networks, and switches alone cannot connect separate networks together, only a router can do that.

(For comparison: if the subnet mask were 255.255.0.0, then the first 16 bits would represent the network, and the remaining bits would be host bits instead.)

Subnetting Example: Dividing a Network into 8 Subnets

Let's say we have the network 192.168.1.0/24, and we want to divide it into 8 separate subnets.

To figure out how many bits we need to borrow from the host portion, we use this formula:

$2^n = 8$, so $n = 3$

This means we need to borrow 3 bits from the host portion. Since we started with a /24 mask, we now add these 3 bits to it: 24 + 3 = 27. So our new subnet mask becomes /27.

In binary, this looks like: 11111111.11111111.11111111.11100000

How many usable host addresses per subnet?

With a /27 mask, we are left with 5 host bits (32 - 27 = 5). Using the formula $2^5 = 32$, we get 32 total addresses per subnet. However, we must subtract 2 addresses from that: one for the network address, and one for the broadcast address, leaving us with 30 usable addresses per subnet.

Why do we remove 2 addresses?

Every subnet needs one address reserved as its network address (representing the subnet itself), and one address reserved as its broadcast address (used to send data to all devices in that subnet at once). These two addresses cannot be assigned to individual devices.

Subnet Table:

Subnet	Network ID	First Usable IP	Last Usable IP	Broadcast IP
1	192.168.1.0	192.168.1.1	192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33	192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65	192.168.1.94	192.168.1.95
4	192.168.1.96	192.168.1.97	192.168.1.126	192.168.1.127
5	192.168.1.128	192.168.1.129	192.168.1.158	192.168.1.159
6	192.168.1.160	192.168.1.161	192.168.1.190	192.168.1.191
7	192.168.1.192	192.168.1.193	192.168.1.222	192.168.1.223
8	192.168.1.224	192.168.1.225	192.168.1.254	192.168.1.255

This table shows how the original network, 192.168.1.0/24, is now divided into 8 smaller, separate subnets, each with its own network ID, usable IP range, and broadcast address.